

```

1)
public class Horse{
    private String name;
    private int weight;

    public Horse(String theName, int theWeight){
        this.name=theName;
        this.weight=theWeight;
    }

    public String getName(){
        return this.name;
    }

    public int getWeight(){
        return this.weight;
    }
}

public class HorseBarn {
    private Horse[] spaces;

    public HorseBarn(int numStalls){
        spaces = new Horse[numStalls];
    }

    public int findHorseSpace(String name) { //The answer starts here!!
        for (int i = 0; i < spaces.length; i++) {
            if (spaces[i] != null && spaces[i].getName().equals(name)) {
                return i;
            }
        }
        return -1;
    }
} //the answer ends here...

```

```

    public static void main (String[]args){
        HorseBarn barn = new HorseBarn(3);
        barn.spaces[0] = new Horse("Trigger", 1340);
        barn.spaces[1] = new Horse("Silver", 1210);
        barn.spaces[2] = new Horse("Duke", 1410);

        System.out.println("Index of Trigger should be 0 and is " +
        barn.findHorseSpace("Trigger"));
        System.out.println("Index of Silver should be 1 and is " + barn.findHorseSpace("Silver"));
        System.out.println("Index of Coco should be -1 and is " + barn.findHorseSpace("Coco"));
    }
}

```

}

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```
Index of Trigger should be 0 and is 0
Index of Silver should be 1 and is 1
Index of Coco should be -1 and is -1
```

2)

```
public class SelfDivisor
```

```
{
```

```
    public static boolean isSelfDivisor(int number){
```

```
        int currNumber = number;
```

```
        int digit = 0;
```

```
        while (currNumber > 0){
```

```
            digit = currNumber%10;
```

```
            if (digit == 0)
```

```
                return false;
```

```
            if(number%digit != 0)
```

```
                return false;
```

```
            currNumber = currNumber/10;
```

```
        }
```

```
        return true;
```

```
}
```

```
public static int[] firstNumSelfDivisors(int start, int num){//answer starts here
```

```
    int[] selfDivisors = new int[num];
```

```
    int count = 0;
```

```
    int current = start;
```

```
    while (count < num){
```

```
        if (isSelfDivisor(current)){
```

```
            selfDivisors[count] = current;
```

```
            count++;
```

```
        }
```

```
        current++;
```

```
    }
```

```

    return selfDivisors;
} //the answer ends here

public static void main (String[] args){
    System.out.println("Self divisors for firstNumSelfDivisors(10,3):");
    for(int n : firstNumSelfDivisors(10,3))
        System.out.print(n + " ");
    System.out.println("Self divisors for firstNumSelfDivisors(22,5):");
    for(int n : firstNumSelfDivisors(22,5))
        System.out.print(n + " ");
}
}

```

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```

Self divisors for firstNumSelfDivisors(10,3):
11 12 15 Self divisors for firstNumSelfDivisors(22,5):
22 24 33 36 44

```

3)

```
import java.util.Arrays;
```

```
public class NumberCube {
```

```

    public int toss() {
        return (int) (Math.random() * 6) + 1;
    }

```

```

    public static int[] getCubeTosses(NumberCube cube, int numTosses) { //answer starts here
        int[] tosses = new int[numTosses];
        for (int i = 0; i < numTosses; i++) {
            tosses[i] = cube.toss();
        }
        return tosses;
    }

```

```
//answer ends here

public static void main(String[] args) {
    NumberCube cube = new NumberCube();
    int numTosses = 9;
    int[] tosses = getCubeTosses(cube, numTosses);

    if (tosses.length < numTosses) {
        System.out.println("It looks like you are not returning an array of the correct size:");
        System.out.println(Arrays.toString(tosses));
    } else {
        System.out.println("You returned an array of the correct size:");
        System.out.println(Arrays.toString(tosses));
    }
}
}
```

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```
You returned an array of the correct size:
[1, 1, 1, 4, 2, 3, 6, 6, 4]

```